

Abstract

The Cognitive Integrity Framework (CIF) [?] establishes formal guarantees for the safe operation of multiagent AI systems through five canonical defense mechanisms: Cognitive Firewalls, Belief Sandboxing, Behavioral Invariants, Drift Detection, and Byzantine Consensus. Paper 2 of this series [?] demonstrated 99.5% detection across 950 attack scenarios. However, the practical applicability of these mechanisms across diverse operational domains has not been systematically evaluated. This paper—Part 4 of the Cognitive Security series—addresses this gap.

We present the **CIF-AD-OODA integration model**, combining CIF's defense mechanisms with Axiomatic Design (AD) theory [?] and Boyd's OODA (Observe-Orient-Decide-Act) Loop [?] to analyze **Goal Hijacking**: an advanced form of indirect prompt injection that constitutes a *teleological attack* on autonomous agency. Unlike conventional exploits targeting data confidentiality, Goal Hijacking corrupts the OODA Loop's **Orientation** phase, overriding an agent's Functional Requirements (FRs) while preserving the appearance of rational behavior. We model these